

IMAT QUESTION BANK: Maths, Physics and Logic (Medium-Hard)

95 questions · 158 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Algebra, functions and logarithms (Q1-16)

1. Given that $\log_{10} 2 \approx 0.30$ and $\log_{10} 3 \approx 0.48$, what is the value of $\log_{10} 7.5$?

A) 0.18 B) 0.58 C) 0.88 D) 1.18 E) 1.88

Answer: C

Given: $\log_{10} 2 \approx 0.30$ and $\log_{10} 3 \approx 0.48$.

Split the argument: $7.5 = 15 \div 2$, so $\log_{10} 7.5 = \log_{10} 3 + \log_{10} 5 - \log_{10} 2$.

Complement rule: $\log_{10} 5 = \log_{10} 10 - \log_{10} 2 = 1 - 0.30 = 0.70$.

Substitute: $\log_{10} 7.5 = 0.48 + 0.70 - 0.30$

Add: $0.48 + 0.70 = 1.18$

Subtract: $1.18 - 0.30 = 0.88$

Result: $\log_{10} 7.5 = 0.88$

Classic slip: 1.18 is $\log_{10} 15$, the value reached by forgetting the division by 2.

2. Solve the equation $\log_3(x + 1) + \log_3(x - 1) = 1$.

A) $x = \pm 2$ B) $x = 2$ only C) $x = -2$ only D) $x = 4$ E) $x = 1$

Answer: B

Given: $\log_3(x + 1) + \log_3(x - 1) = 1$

Product rule: $\log_3((x + 1)(x - 1)) = \log_3(x^2 - 1)$, so $\log_3(x^2 - 1) = 1$.

Exponent form: $x^2 - 1 = 3$

Add 1: $x^2 = 4$, so $x = +2$ or $x = -2$

Domain: both arguments need $x > 1$, so $x = -2$ makes $\log_3(x - 1)$ undefined and is rejected.

Result: $x = 2$ only

Classic slip: keeping $x = \pm 2$ ignores the domain check at $x = -2$.

3. What is the value of $\log_8 32$?

A) 2 B) $3/2$ C) 3 D) $4/3$ E) $5/3$

Answer: E

Given: $\log_8 32$.

Change of base: $\log_8 32 = \log_2 32 \div \log_2 8$.

Powers of 2: $32 = 2^5$ so $\log_2 32 = 5$, and $8 = 2^3$ so $\log_2 8 = 3$

Substitute: $\log_8 32 = 5 \div 3$

Result: $5/3$.

Classic slip: $3/2$ inverts the ratio by dividing $\log_2 8$ by $\log_2 32$.

4. What is the solution of $\log_2 x + \log_4 x = 6$?

- A) 4 B) 8 C) 16 D) 32 E) 64

Answer: C

Given: $\log_2 x + \log_4 x = 6$

Base relation: $\log_4 x = \log_2 x \div \log_2 4 = (1/2) \log_2 x$, so the equation reads $\log_2 x + (1/2) \log_2 x = 6$.

Combine like terms: $(3/2) \log_2 x = 6$

Solve: $\log_2 x = 6 \times 2 \div 3 = 4$

Exponent form: $x = 2^4 = 16$

Check: $\log_2 16 + \log_4 16 = 4 + 2 = 6$

Result: $x = 16$

Classic slip: 64 solves $\log_2 x = 6$ alone and drops the $\log_4$ term.

5. Which interval is the solution set of $(x - 1)(x + 2)(x - 4) > 0$?

- A) $-2 < x < 4$
B) $1 < x < 4$
C) $x < -2$ or $x > 4$
D) $-2 < x < 1$ or $x > 4$
E) $x < -2$ or $1 < x < 4$

Answer: D

Given: $(x - 1)(x + 2)(x - 4) > 0$ with roots $-2, 1, 4$; test one point per interval.

$x = -3$ (below -2): $(-4)(-1)(-7) = -28$, negative, fails

$x = 0$ (between -2 and 1): $(-1)(2)(-4) = 8$, positive, holds

$x = 2$ (between 1 and 4): $(1)(4)(-2) = -8$, negative, fails

$x = 5$ (above 4): $(4)(7)(1) = 28$, positive, holds

Result: $-2 < x < 1$ or $x > 4$.

Classic slip: $x < -2$ or $x > 4$ tracks only the outer factors and ignores the middle root at $x = 1$.

6. What is the solution set of the inequality $(x - 3)/(x + 2) \leq 0$?

- A) $-2 < x \leq 3$
B) $x \leq 3$
C) $x < -2$ or $x \geq 3$
D) $-2 \leq x \leq 3$
E) $-2 < x < 3$

Answer: A

Given: $(x - 3)/(x + 2) \leq 0$.

Critical points: the fraction is 0 at $x = 3$ and undefined at $x = -2$

Test $x = 0$, between them: $(0 - 3)/(0 + 2) = -3/2$, negative, holds.

Test $x = -3$: $(-5)/(-1) = 5$, positive, fails; test $x = 4$: $1/6$, positive, fails.

So the fraction is negative exactly for $-2 < x < 3$; $x = 3$ joins through the $\leq$ zero, and $x = -2$ is excluded as undefined.

Result: $-2 < x \leq 3$.

Classic slip: $x \leq 3$ ignores the sign flip at $x = -2$, where the fraction is positive on both sides.

7. What is the solution set of $|2x - 5| < 7$?

- A) $1 < x < 6$ B) $-6 < x < -1$ C) $-6 < x < 6$ D) $-1 < x < 6$ E) $x < 6$

Answer: D

Given: $|2x - 5| < 7$.

Absolute value rule: $-7 < 2x - 5 < 7$.

Add 5 to all three parts: $-7 + 5 = -2$ and $7 + 5 = 12$, so $-2 < 2x < 12$

Divide by 2: $-1 < x < 6$.

Result: $-1 < x < 6$.

Classic slip: $1 < x < 6$ keeps only the branch $2x - 5 < 7$ and drops the branch $2x - 5 > -7$.

8. What is the solution set of the inequality $x^2 < x$?

- A) $0 < x < 1$ B) $x < 0$ C) $x < 0$ or $x > 1$ D) $x < 1$ E) $x > 1$

Answer: A

Given: $x^2 < x$.

Rearrange: $x^2 - x < 0$, that is $x(x - 1) < 0$.

Roots: $x = 0$ and $x = 1$; a product of two factors is negative exactly when the factors have opposite signs.

Test $x = 1/2$: $(1/2) \times (-1/2) = -1/4$, negative, holds; for $x < 0$ both factors are negative and the product is positive, failing.

Result: $0 < x < 1$.

Classic slip: dividing by x to reach $x < 1$ assumes x is positive, but every negative x makes the left side positive and the right side negative.

9. Which expression is equal to $\sqrt{7 + 4\sqrt{3}}$?

- A) $4 + \sqrt{3}$ B) $2 + \sqrt{3}$ C) $\sqrt{7} + \sqrt{3}$ D) $2\sqrt{3} + 1$ E) $1 + \sqrt{6}$

Answer: B

Given: $\sqrt{7 + 4\sqrt{3}}$.

Square pattern: $a^2 + 2ab + b^2$ needs $a^2 + b^2 = 7$ and $2ab = 4\sqrt{3}$

Try $a = 2$, $b = \sqrt{3}$: $a^2 = 4$, $b^2 = 3$, $2ab = 2 \times 2 \times \sqrt{3} = 4\sqrt{3}$

Sum: $4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$, matching the radicand exactly

So $7 + 4\sqrt{3} = (2 + \sqrt{3})^2$, and since $2 + \sqrt{3} > 0$ the square root is $2 + \sqrt{3}$

Result: $2 + \sqrt{3}$.

Classic slip: $1 + \sqrt{6}$ squares to $7 + 2\sqrt{6}$, whose radical term does not match $4\sqrt{3}$.

10. What is the value of $27^{(2/3)} + 16^{(3/4)}$?

- A) 13 B) 17 C) 21 D) 26 E) 30

Answer: B

Given: $27^{(2/3)} + 16^{(3/4)}$.

Rule: $x^{(m/n)} = (\sqrt[n]{x})^m$, take the root first, then the power.

$27^{(2/3)}$: $\sqrt[3]{27} = 3$, then $3^2 = 9$

$16^{(3/4)}$: $\sqrt[4]{16} = 2$, then $2^3 = 8$

Add: $9 + 8 = 17$

Result: 17.

Classic slip: 30 multiplies base by exponent instead, $27 \times 2/3 + 16 \times 3/4 = 18 + 12$.

11. If $2^a = 3$ and $2^b = 5$, which of the following is equal to 60?

- A) $2^{(a+b)}$ B) $2^{(2a+b)}$ C) $2^{(a+b+2)}$ D) $2^{(a+2b)}$ E) $2^{(ab)}$

Answer: C

Given: $2^a = 3$ and $2^b = 5$

Prime split: $60 = 4 \times 3 \times 5$

Replace 3 and 5: $60 = 2^2 \times 2^a \times 2^b$

Product rule, exponents add: $2^{(2+a+b)}$.

Result: $2^{(a+b+2)}$.

Classic slip: $2^{(ab)}$ multiplies the exponents, which is the rule for $(2^a)^b$, not for a product of powers.

12. For a school show, adult tickets cost 12 euro and child tickets cost 7 euro. In total 180 tickets were sold, and the takings were 1730 euro. How many adult tickets were sold?

- A) 84 B) 86 C) 90 D) 94 E) 96

Answer: D

Given: $a + c = 180$ and $12a + 7c = 1730$ euro, with a adult and c child tickets.

Substitute $c = 180 - a$: $12a + 7(180 - a) = 1730$

Expand: $12a + 1260 - 7a = 1730$, so $5a = 1730 - 1260 = 470$

Divide: $a = 470 \div 5 = 94$

Check: $94 \times 12 + 86 \times 7 = 1128 + 602 = 1730$

Result: 94 adult tickets.

Classic slip: 86 is the number of child tickets, from stopping at c instead of a .

13. The sum of two numbers is 45 and their difference is 13. What is the larger number?

- A) 13 B) 16 C) 29 D) 32 E) 45

Answer: C

Given: $x + y = 45$ and $x - y = 13$, with x the larger number

Add the equations: $(x + y) + (x - y) = 2x = 45 + 13 = 58$

Divide: $x = 58 \div 2 = 29$

Then $y = 45 - 29 = 16$, and the check $29 - 16 = 13$ holds

Result: the larger number is 29.

Classic slip: $32 = 45 - 13$ is twice the smaller number, from subtracting the equations instead of adding.

14. A laboratory technician mixes a 20% saline solution with a 50% saline solution to obtain 6 dm³ of a 30% saline solution. What volume of the 50% solution is needed?

- A)** 1 dm³ **B)** 2 dm³ **C)** 2.5 dm³ **D)** 3 dm³ **E)** 4 dm³

Answer: B

Given: 6 dm³ of 30% saline mixed from a 20% and a 50% solution; let y dm³ be the 50% part.

Salt balance: $0.20(6 - y) + 0.50y = 0.30 \times 6$

Compute the target salt: $0.30 \times 6 = 1.8$, and expand $0.20(6 - y) = 1.2 - 0.2y$

So $1.2 - 0.2y + 0.5y = 1.8$, that is $1.2 + 0.3y = 1.8$, so $0.3y = 0.6$

Divide: $y = 0.6 \div 0.3 = 2$ dm³, and the 20% part is then 4 dm³

Result: 2 dm³ of the 50% solution.

Classic slip: 4 dm³ is the volume of the 20% solution.

15. The function f is defined by $f(x) = (x + 1)/(x - 1)$ for $x \neq 1$. What is $f(f(3))$?

- A)** 1/2 **B)** 2 **C)** 3 **D)** 4 **E)** 9

Answer: C

Given: $f(x) = (x + 1)/(x - 1)$; find $f(f(3))$

Inner step: $f(3) = (3 + 1)/(3 - 1) = 4/2 = 2$

Outer step: $f(2) = (2 + 1)/(2 - 1) = 3/1 = 3$

So $f(f(3)) = 3$; this function is its own inverse.

Result: 3.

Classic slip: 2 stops after a single application of f .

16. The function f is defined by $f(x) = (2x + 1)/(x - 3)$ for $x \neq 3$. Which of the following is the inverse function $f^{-1}(x)$?

- A)** $(3x - 1)/(x + 2)$
B) $(x + 3)/(2x - 1)$
C) $(2x + 1)/(x - 3)$
D) $(3x + 1)/(x + 2)$
E) $(3x + 1)/(x - 2)$

Answer: E

Given: $f(x) = (2x + 1)/(x - 3)$

Set $y = (2x + 1)/(x - 3)$ and solve for x

Multiply through: $y(x - 3) = 2x + 1$, so $xy - 3y = 2x + 1$

Collect the x terms: $xy - 2x = 3y + 1$, so $x(y - 2) = 3y + 1$

Divide: $x = (3y + 1)/(y - 2)$, hence $f^{-1}(x) = (3x + 1)/(x - 2)$

Check: $f(4) = 9/1 = 9$ and $f^{-1}(9) = (27 + 1)/(9 - 2) = 28/7 = 4$

Result: $(3x + 1)/(x - 2)$.

Classic slip: $(2x + 1)/(x - 3)$ is f itself, left unchanged instead of inverted.

Section B - Geometry, trigonometry and graphs (Q17-28)

17. A solid wooden block has the shape of a right circular cylinder of radius 6 cm and height 9 cm. A conical hole with the same base radius and the same height is drilled out of the block. What fraction of the cylinder's volume remains?

- A)** $1/3$ **B)** $1/2$ **C)** $2/3$ **D)** $3/4$ **E)** $8/9$

Answer: C

Given: cylinder radius 6 cm, height 9 cm; conical hole with the same base and height.

Formulas: $V_{\text{cylinder}} = \pi r^2 h$ and $V_{\text{cone}} = (1/3)\pi r^2 h$

With the same r and h the cone is exactly one third of the cylinder's volume, so the 6 and 9 cancel.

Remaining fraction: $1 - 1/3 = 2/3$

Result: $2/3$ of the volume remains.

Classic slip: $1/3$ is the fraction drilled out, not the fraction left.

18. A circle of radius 5 cm is inscribed in a square. Taking $\pi = 3.14$, what is the area of the part of the square that lies outside the circle?

- A)** 21.5 cm^2 **B)** 39.25 cm^2 **C)** 78.5 cm^2 **D)** 100 cm^2 **E)** 121.5 cm^2

Answer: A

Given: circle radius 5 cm inscribed in a square, $\pi = 3.14$.

An inscribed circle touches all four sides, so the square side = $2 \times 5 = 10 \text{ cm}$.

Square area: $10 \times 10 = 100 \text{ cm}^2$

Circle area: $3.14 \times 5^2 = 3.14 \times 25 = 78.5 \text{ cm}^2$

Subtract: $100 - 78.5 = 21.5 \text{ cm}^2$

Result: 21.5 cm^2 .

Classic slip: 121.5 cm^2 adds the two areas instead of subtracting them.

19. An equilateral triangle has sides of length 12 cm. What is its area?

- A)** $36\sqrt{3} \text{ cm}^2$ **B)** 72 cm^2 **C)** $72\sqrt{3} \text{ cm}^2$ **D)** 144 cm^2 **E)** $144\sqrt{3} \text{ cm}^2$

Answer: A

Given: equilateral triangle, side 12 cm.

Drop a height: it bisects the base into 6 cm and 6 cm, giving a right triangle with hypotenuse 12 cm and short leg 6 cm.

Height: $h = \sqrt{12^2 - 6^2} = \sqrt{144 - 36} = \sqrt{108} = 6\sqrt{3} \text{ cm}$

Area: $(1/2) \times \text{base} \times \text{height} = (1/2) \times 12 \times 6\sqrt{3} = 6 \times 6\sqrt{3} = 36\sqrt{3} \text{ cm}^2$

Result: $36\sqrt{3} \text{ cm}^2$.

Classic slip: 72 cm^2 uses the side length 12 as if it were the height.

20. A closed cylindrical tin has radius 3 cm and height 8 cm. What is its total surface area, ends included?

- A)** $42\pi \text{ cm}^2$ **B)** $48\pi \text{ cm}^2$ **C)** $57\pi \text{ cm}^2$ **D)** $66\pi \text{ cm}^2$ **E)** $72\pi \text{ cm}^2$

Answer: D

Given: closed cylinder, radius 3 cm, height 8 cm.

Formula: total surface = two ends + curved wall = $2\pi r^2 + 2\pi rh$.

Ends: $2\pi \times 3^2 = 2\pi \times 9 = 18\pi \text{ cm}^2$

Wall: $2\pi \times 3 \times 8 = 2\pi \times 24 = 48\pi \text{ cm}^2$

Add: $18\pi + 48\pi = 66\pi \text{ cm}^2$

Result: $66\pi \text{ cm}^2$.

Classic slip: $48\pi \text{ cm}^2$ counts only the wall, and $57\pi \text{ cm}^2$ forgets one of the two ends.

21. What is the angle between the hour hand and the minute hand of a clock at 3:40? (The hour hand turns at 0.5° per minute, the minute hand at 6° per minute.)

- A)** 100° **B)** 110° **C)** 130° **D)** 150° **E)** 230°

Answer: C

Given: 3:40; hour hand 0.5° per minute, minute hand 6° per minute, angles measured from 12.

Hour hand: $3 \times 30^\circ + 40 \times 0.5^\circ = 90^\circ + 20^\circ = 110^\circ$

Minute hand: $40 \times 6^\circ = 240^\circ$

Angle between: $240^\circ - 110^\circ = 130^\circ$

Result: 130° .

Classic slip: 150° freezes the hour hand at exactly 3 o'clock and ignores its 20° drift during the 40 minutes.

22. Between 7:00 and 8:00 the hands of a clock are exactly 100 degrees apart at 7:20. At what later time in that hour are they exactly 100 degrees apart again? (The hour hand turns at 0.5 degrees per minute, the minute hand at 6 degrees per minute.)

- A)** 7:52 and 8/11 minutes
B) 7:54 and 6/11 minutes
C) 7:56 and 4/11 minutes
D) 7:58 and 2/11 minutes
E) 8:00 exactly

Answer: C

Given: m minutes past 7:00; hour hand at $210^\circ + 0.5m$, minute hand at $6m$, both measured from 12.

Separation: $|6m - (210 + 0.5m)| = |5.5m - 210|$ degrees

The first solution, $5.5m = 110$, gives $m = 20$, the given 7:20

The later solution: $5.5m - 210 = 100$, so $5.5m = 310$

Divide: $m = 310 \div 5.5 = 56$ and $4/11$ minutes, since $310/5.5 = 620/11$

Result: 7:56 and $4/11$ minutes.

Classic slip: 8:00 exactly assumes the hands close at 5° per minute instead of the true $6 - 0.5 = 5.5^\circ$ per minute.

23. What is the value of $\tan 30^\circ \times \tan 60^\circ$?

- A) 3 B) $\sqrt{3}/3$ C) $\sqrt{3}$ D) $1/3$ E) 1

Answer: E

Given: $\tan 30^\circ \times \tan 60^\circ$.

Values: $\tan 30^\circ = \sqrt{3}/3$ and $\tan 60^\circ = \sqrt{3}$

Substitute: $(\sqrt{3}/3) \times \sqrt{3} = (\sqrt{3} \times \sqrt{3})/3 = 3/3 = 1$

Check: 30° and 60° are complementary, and complementary angles have reciprocal tangents, so the product must be 1.

Result: 1.

Classic slip: $1/3$ uses $\tan 30^\circ$ twice, and 3 divides the two values instead of multiplying them.

24. For an angle θ , what is the value of $(\sin \theta + \cos \theta)^2 - 2 \sin \theta \cos \theta$?

- A) $\sin^2 \theta - \cos^2 \theta$ B) 2 C) 0 D) $2 \sin \theta \cos \theta$ E) 1

Answer: E

Given: $(\sin \theta + \cos \theta)^2 - 2 \sin \theta \cos \theta$.

Expand the square: $\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta$.

Subtract the middle term: the $+2 \sin \theta \cos \theta$ and the $-2 \sin \theta \cos \theta$ cancel exactly.

What remains: $\sin^2 \theta + \cos^2 \theta = 1$ by the Pythagorean identity.

Result: 1.

Classic slip: $2 \sin \theta \cos \theta$ stops after the expansion and forgets the cancellation.

25. What is the value of $(\sin 45^\circ - \cos 45^\circ)^2 + 2 \sin 30^\circ \cos 60^\circ$?

- A) 0 B) $1/4$ C) $1/2$ D) 1 E) $3/4$

Answer: C

Given: $(\sin 45^\circ - \cos 45^\circ)^2 + 2 \sin 30^\circ \cos 60^\circ$.

Values: $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$, $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$.

First term: $(\sqrt{2}/2 - \sqrt{2}/2)^2 = 0^2 = 0$

Second term: $2 \times (1/2) \times (1/2) = 1 \times (1/2) = 1/2$

Add: $0 + 1/2 = 1/2$

Result: $1/2$.

Classic slip: 0 forgets the second term altogether.

26. A straight line passes through the points (2, 5) and (6, 17). What is its slope?

- A) $1/4$ B) $1/3$ C) 3 D) 4 E) 12

Answer: C

Given: points (2, 5) and (6, 17).

Formula: slope = change in y $\div$ change in x.

Substitute: $(17 - 5) \div (6 - 2)$.

Numerator: $17 - 5 = 12$; denominator: $6 - 2 = 4$

Divide: $12 \div 4 = 3$

Result: 3.

Classic slip: 4 and 12 each use only one of the two changes, and $1/3$ inverts the ratio.

27. A cyclist rides along a straight road at a constant speed. Her distance from the start is recorded as 1.8 km after 6 minutes and 3.0 km after 10 minutes. What is her speed in m/s?

- A)** 3 m/s **B)** 5 m/s **C)** 18 m/s **D)** 50 m/s **E)** 300 m/s

Answer: B

Given: 1.8 km from the start at 6 min, 3.0 km at 10 min, constant speed.

Between the records: distance $3.0 - 1.8 = 1.2$ km = 1200 m; time $10 - 6 = 4$ min = 240 s.

Speed = distance $\div$ time = $1200 \text{ m} \div 240 \text{ s}$

Divide: $1200 \div 240 = 5$ m/s

Result: 5 m/s.

Classic slip: 18 m/s is the speed in km/h (0.3 km per minute $\times 60$) left unconverted, and 300 m/s keeps the rate in metres per minute.

28. Water is pumped into a tank that already contains 40 L. The volume reaches 90 L after 5 minutes and 160 L after 15 minutes. What is the average rate of inflow over the whole 15 minutes?

- A)** 7 L/min **B)** 8 L/min **C)** 8.5 L/min **D)** 10 L/min **E)** 160 L/min

Answer: B

Given: 40 L at the start, 90 L at 5 min, 160 L at 15 min.

Average rate = total change $\div$ total time = $(160 - 40) \text{ L} \div 15 \text{ min}$.

Change: $160 - 40 = 120$ L

Divide: $120 \div 15 = 8$ L/min

Result: 8 L/min.

Classic slip: 8.5 L/min averages the two segment rates (10 L/min then 7 L/min) instead of dividing the total change by the total time.

Section C - Probability and combinatorics (Q29-36)

29. Two fair six-sided dice are rolled once. What is the probability that the sum of the two scores is at least 9?

- A)** $1/9$ **B)** $1/3$ **C)** $5/36$ **D)** $5/18$ **E)** $5/12$

Answer: D

Given: two fair dice, 36 ordered outcomes; want a sum of at least 9.

Count per sum: sum 9 in 4 ways, sum 10 in 3, sum 11 in 2, sum 12 in 1.

Total favourable: $4 + 3 + 2 + 1 = 10$

$P = 10/36 = 5/18$

Result: $5/18$.

Classic slip: $5/36$ counts only the five unordered pairs, forgetting that (4,6) and (6,4) are different rolls, and $1/9$ answers a sum of exactly 9.

30. Three fair six-sided dice are rolled. What is the probability that all three show the same number?

- A)** $1/216$ **B)** $1/108$ **C)** $1/72$ **D)** $1/36$ **E)** $1/6$

Answer: D

Given: three fair dice, 216 ordered outcomes; want all three equal.

Chain the conditions: the first die shows anything (probability 1), the second must match it ($1/6$), the third must match too ($1/6$).

Multiply: $1 \times (1/6) \times (1/6) = 1/36$

Cross-check: 6 matching triples out of 216 outcomes give $6/216 = 1/36$.

Result: $1/36$.

Classic slip: $1/216$ fixes all three faces in advance, counting one named triple instead of all six, and $1/6$ is three copies of one particular named number.

31. Two fair six-sided dice are rolled. What is the probability that the two scores are a 2 and a 5, in either order?

- A)** $1/36$ **B)** $1/18$ **C)** $1/9$ **D)** $1/6$ **E)** $1/3$

Answer: B

Given: two fair dice, 36 ordered outcomes; want a 2 and a 5 in either order.

Favourable ordered outcomes: (2,5) and (5,2), that is 2 of the 36.

$P = 2/36 = 1/18$

Result: $1/18$.

Classic slip: $1/36$ takes a single ordering, and $1/9$ counts four outcomes by also including the doubles (2,2) and (5,5).

32. Two fair six-sided dice are rolled. What is the probability that the sum is 7 or at least one die shows a six?

- A)** $1/6$ **B)** $5/12$ **C)** $7/18$ **D)** $11/36$ **E)** $17/36$

Answer: B

Given: two fair dice, 36 ordered outcomes; event A is a sum of 7, event B is at least one six.

$P(A) = 6/36$; $P(B) = 11/36$, since 36 outcomes minus 25 with no six.

Overlap: A and B share (1,6) and (6,1), that is 2 outcomes.

Inclusion-exclusion: $(6 + 11 - 2)/36 = 15/36$

Reduce: $15/36 = 5/12$

Result: $5/12$.

Classic slip: $17/36$ adds the two probabilities without subtracting the overlap.

33. A fair coin is flipped four times. What is the probability of getting exactly two heads?

- A)** $1/8$ **B)** $1/4$ **C)** $1/2$ **D)** $3/16$ **E)** $3/8$

Answer: E

Given: 4 fair flips, $2^4 = 16$ equally likely head-tail sequences; want exactly two heads.

Count the arrangements of two heads among four flips: $C(4,2) = (4 \times 3) \div (2 \times 1) = 6$.

$P = 6/16 = 3/8$

Result: $3/8$.

Classic slip: $1/4$ counts only a few of the six arrangements instead of all of them.

34. An urn contains 3 red, 4 green and 5 blue balls. Two balls are drawn at random without replacement. What is the probability that at least one of them is green?

- A)** $1/3$ **B)** $3/4$ **C)** $14/33$ **D)** $16/33$ **E)** $19/33$

Answer: E

Given: urn with 3 red, 4 green, 5 blue balls (12 total); draw 2 without replacement; want at least one green.

Complement route: no green means both draws come from the 8 non-green balls.

$$P(\text{no green}) = (8/12) \times (7/11) = 56/132 = 14/33$$

$$P(\text{at least one green}) = 1 - 14/33 = 19/33$$

Result: $19/33$.

Classic slip: $16/33$ answers exactly one green, and $14/33$ is the complement itself.

35. The probability that a candidate passes a maths test is $2/3$, and the probability that she passes an independent logic test is $3/4$. What is the probability that she passes exactly one of the two tests?

- A)** $1/2$ **B)** $1/6$ **C)** $1/12$ **D)** $5/12$ **E)** $11/12$

Answer: D

Given: $P(\text{maths pass}) = 2/3$, $P(\text{logic pass}) = 3/4$, independent; want exactly one pass.

Case 1, pass maths and fail logic: $(2/3) \times (1/4) = 2/12$.

Case 2, fail maths and pass logic: $(1/3) \times (3/4) = 3/12$.

Add the disjoint cases: $2/12 + 3/12 = 5/12$

Result: $5/12$.

Classic slip: $1/2$ is the probability of passing both, and $11/12$ is the probability of passing at least one.

36. A fair coin is tossed and a fair six-sided die is rolled. What is the probability of obtaining tails and a score greater than 4?

- A)** $1/12$ **B)** $1/6$ **C)** $1/4$ **D)** $1/2$ **E)** $2/3$

Answer: B

Given: fair coin and fair die, independent; want tails and a score above 4.

$$P(\text{tails}) = 1/2; P(\text{score above 4}) = 2/6 = 1/3, \text{ the faces 5 and 6}$$

$$\text{Independent events multiply: } (1/2) \times (1/3) = 1/6$$

Result: $1/6$.

Classic slip: $2/3$ computes the union of the two events instead of their joint occurrence, and $1/12$ uses a single high face instead of two.

Section D - Unit conversion and arithmetic (Q37-40)

37. A high-speed train covers 270 km in 3.0 h at a constant speed. Expressed in m/s, its speed is:

- A) 25 m/s B) 75 m/s C) 90 m/s D) 324 m/s E) 1500 m/s

Answer: A

Given: 270 km in 3.0 h at constant speed.

Speed: $270 \div 3.0 = 90$ km/h

Convert: 90 km/h = 90 000 m per 3600 s

Divide: $90\,000 \div 3600 = 25$ m/s

Result: 25 m/s.

Classic slip: 324 m/s multiplies 90 by 3.6 instead of dividing, and 1500 m/s gives metres per minute rather than metres per second.

38. A swimming pool is 5.0 m long, 2.0 m wide and 1.2 m deep. What volume of water does it hold, expressed in litres?

- A) 12 L B) 120 L C) 1200 L D) 12000 L E) 120000 L

Answer: D

Given: pool 5.0 m long, 2.0 m wide, 1.2 m deep.

Volume: $5.0 \times 2.0 = 10$ m², then $10 \times 1.2 = 12$ m³

Convert: 1 m³ = 1000 dm³ and 1 dm³ = 1 L, so $12 \times 1000 = 12\,000$ L

Result: 12 000 L.

Classic slip: 1200 L multiplies by 100 instead of 1000, using a length factor for a volume.

39. A water tank is $\frac{2}{3}$ full. A valve then drains $\frac{1}{4}$ of the tank's total capacity. What fraction of the tank is still full?

- A) $\frac{1}{6}$ B) $\frac{1}{4}$ C) $\frac{1}{2}$ D) $\frac{5}{12}$ E) $\frac{11}{12}$

Answer: D

Given: tank $\frac{2}{3}$ full; the valve drains $\frac{1}{4}$ of the total capacity.

Remaining fraction: $\frac{2}{3} - \frac{1}{4}$.

Common denominator 12: $\frac{8}{12} - \frac{3}{12} = \frac{5}{12}$

Result: $\frac{5}{12}$ of the tank is still full.

Classic slip: $\frac{1}{2}$ instead removes a quarter of the water already present ($\frac{2}{3} \times \frac{3}{4}$), and $\frac{11}{12}$ adds the fractions instead of subtracting.

40. A water tank is one third full. After 40 more litres are poured in, the tank is three quarters full. What is the full capacity of the tank?

- A)** 60 litres **B)** 72 litres **C)** 90 litres **D)** 96 litres **E)** 120 litres

Answer: D

Given: tank $\frac{1}{3}$ full; after 40 L poured in it is $\frac{3}{4}$ full.

Fraction filled by the 40 L: $\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$ of the capacity

So $(\frac{5}{12}) \times \text{capacity} = 40 \text{ L}$, and $\text{capacity} = 40 \times \frac{12}{5}$

Compute: $40 \times 12 = 480$, then $480 \div 5 = 96 \text{ L}$

Result: 96 L.

Classic slip: 120 L treats the 40 L as one third of the tank, and 60 L treats it as the two thirds needed to reach completely full.

Section E - Kinematics and dynamics (Q41-56)

41. The velocity of a trolley moving along a straight track is recorded every 2 s: at 0 s it is 0 m/s, at 2 s it is 6 m/s, at 4 s it is 10 m/s, at 6 s it is 12 m/s and at 8 s it is 12 m/s. Which statement is correct?

- A)** Its acceleration between 4 s and 6 s is 1 m/s^2 .
B) Its average velocity over the whole 8 s is 10 m/s.
C) It decelerates between 6 s and 8 s.
D) Its acceleration is constant at 3 m/s^2 between 0 s and 6 s.
E) It travels 96 m during the 8 s.

Answer: A

Given: $v = 0, 6, 10, 12, 12 \text{ m/s}$ at $t = 0, 2, 4, 6, 8 \text{ s}$

Segment accelerations: $(6 - 0)/2 = 3$, $(10 - 6)/2 = 2$, $(12 - 10)/2 = 1$, $(12 - 12)/2 = 0 \text{ m/s}^2$, so the claim of a constant 3 m/s^2 between 0 s and 6 s fails.

Displacement, trapezium area per 2 s block: $(0 + 6)/2 \times 2 = 6 \text{ m}$, $(6 + 10)/2 \times 2 = 16 \text{ m}$, $(10 + 12)/2 \times 2 = 22 \text{ m}$, $12 \times 2 = 24 \text{ m}$

Total: $6 + 16 + 22 + 24 = 68 \text{ m}$, so the average velocity is $68 \div 8 = 8.5 \text{ m/s}$, not 10 m/s, and the distance is not 96 m.

From 6 s to 8 s the velocity is constant at 12 m/s, so there is no deceleration.

Result: the acceleration between 4 s and 6 s is $(12 - 10) \div 2 = 1 \text{ m/s}^2$, the only correct claim.

Classic slip: reading a constant 3 m/s^2 from the first segment, or multiplying the final 12 m/s by 8 s to get 96 m.

42. In a race, a sprinter increases her speed steadily from 0 to 8.0 m/s during the first 4.0 s and then keeps running at a constant 8.0 m/s for the next 6.0 s. What is her average speed over the whole 10 s?

- A)** 1.6 m/s **B)** 4.0 m/s **C)** 4.8 m/s **D)** 6.4 m/s **E)** 8.0 m/s

Answer: D

Given: 0 to 8.0 m/s uniformly in 4.0 s, then 8.0 m/s constant for 6.0 s.

Distance is the area under the velocity-time graph.

Phase 1, triangle: $(1/2) \times 4.0 \times 8.0 = 16$ m

Phase 2, rectangle: $8.0 \times 6.0 = 48$ m

Total: $16 + 48 = 64$ m

Average speed: $64 \div 10 = 6.4$ m/s

Result: 6.4 m/s.

Classic slip: 8.0 m/s takes the final speed as the whole-run average, and 4.0 m/s would hold only if the acceleration lasted all 10 s.

43. On a straight air track, a glider passes sensor P moving at 2.0 m/s and sensor Q moving at 4.0 m/s. The sensors are 6.0 m apart and the glider takes 2.0 s to travel from P to Q. Which statement is correct?

- A)** Its average velocity between the gates is 3.0 m/s and its acceleration is 2.0 m/s².
B) Its average velocity between the gates is 2.0 m/s and its acceleration is 1.0 m/s².
C) Its average velocity between the gates is 4.0 m/s and its acceleration is 1.0 m/s².
D) Its average velocity between the gates is 6.0 m/s and its acceleration is 1.0 m/s².
E) Its average velocity between the gates is 3.0 m/s and its acceleration is 1.0 m/s².

Answer: E

Given: $v_P = 2.0$ m/s, $v_Q = 4.0$ m/s, separation 6.0 m, travel time 2.0 s

Average velocity = displacement $\div$ time = $6.0 \div 2.0 = 3.0$ m/s

Acceleration = change in velocity $\div$ time = $(4.0 - 2.0) \div 2.0 = 2.0 \div 2.0 = 1.0$ m/s²

Result: 3.0 m/s average velocity and 1.0 m/s² acceleration.

Classic slip: 2.0 m/s² divides the speed change by 1 s instead of 2 s, and 2.0 or 4.0 m/s borrow one instantaneous gate speed for the whole run.

44. Which of the following statements about motion graphs is correct?

- A)** For uniformly accelerated motion starting from rest, the average velocity during the acceleration is half the final velocity.
- B)** A curved velocity-time graph means the object is moving along a circular path.
- C)** The average velocity over any time interval always equals the mean of the initial and final velocities.
- D)** The area under a velocity-time graph gives the acceleration.
- E)** The gradient of a velocity-time graph gives the distance travelled.

Answer: A

Givens: five claims about motion graphs.

Graph rules: the gradient of a velocity-time graph gives acceleration, its area gives displacement, and the mean of the end velocities equals the average only under uniform acceleration.

Test the remaining claim: for uniform acceleration from rest, $v = at$, so the average velocity is $(0 + at) \div 2 = (1/2)at$.

The final velocity is at , so the average is exactly half of it.

Result: for uniformly accelerated motion starting from rest, the average velocity during the acceleration is half the final velocity.

Classic slip: the claims reading distance from the gradient, or acceleration from the area, swap the two graph readings.

45. Which statement correctly distinguishes the tangential and the centripetal components of acceleration for an object moving in a circle?

- A)** The tangential component acts towards the centre, and the centripetal component acts along the tangent.
- B)** Both components change only the direction of motion.
- C)** Both components change only the speed of the object.
- D)** The centripetal component changes the speed, and the tangential component changes the direction of motion.
- E)** The tangential component changes the speed, and the centripetal component changes the direction of the velocity.

Answer: E

Givens: two acceleration components for an object moving in a circle.

Tangential component: acts along the tangent, changes the magnitude of the velocity, that is the speed.

Centripetal component: acts towards the centre, changes only the direction of the velocity.

Matching roles to names gives the correct statement.

Result: the tangential component changes the speed, and the centripetal component changes the direction of the velocity.

Classic slip: the reversed version swaps the two roles, and claims that both components act on speed alone or direction alone contradict the definitions.

46. A cyclist rides around a circular track of radius 20 m at a constant speed of 10 m/s. Which statement about her acceleration is correct?

- A)** It is 5.0 m/s², directed towards the centre of the track.
- B)** It is 2.0 m/s², directed away from the centre of the track.
- C)** It is 5.0 m/s², directed along her direction of travel.
- D)** It is 0.50 m/s², directed along her direction of travel.
- E)** It is zero, because her speed is constant.

Answer: A

Given: $r = 20$ m, constant speed 10 m/s

Constant speed on a circle still means a continuously changing direction, so the acceleration is purely centripetal: $a = v^2/r$.

Substitute: $a = 10^2 \div 20 = 100 \div 20 = 5.0$ m/s²

Direction: towards the centre.

Result: 5.0 m/s² towards the centre.

Classic slip: 0.50 m/s² and 2.0 m/s² come from v/r and r/v instead of v^2/r , and zero acceleration ignores the direction change.

47. A car is driven around a circular test track while the driver steadily presses the accelerator, so the car is speeding up. At such an instant, which statement about the car's acceleration is correct?

- A)** It has a component along the motion and a component towards the centre, so the resultant points forwards and inwards.
- B)** It points directly towards the centre of the track.
- C)** It is zero, because the tangential and centripetal parts cancel.
- D)** It points directly away from the centre, in the direction of the outward push the driver feels.
- E)** It points along the direction of motion, since only a change of speed can produce acceleration.

Answer: A

Givens: car on a circular track with the speed steadily increasing.

A changing speed requires a tangential component along the direction of motion.

A changing direction requires a centripetal component towards the centre.

The resultant of a forward and an inward component points forwards and inwards, between tangent and radius.

Result: the acceleration has a component along the motion and a component towards the centre, so the resultant points forwards and inwards.

Classic slip: pointing straight at the centre would require constant speed, and a resultant pointing outwards never occurs in circular motion.

48. A car travels at 72 km/h along a straight road when the driver sees a hazard. Her reaction time is 1.0 s, after which the brakes produce a constant deceleration of 5.0 m/s² until the car stops. What total distance does the car cover between the hazard being seen and the car stopping?

- A)** 20 m **B)** 40 m **C)** 60 m **D)** 80 m **E)** 100 m

Answer: C

Given: 72 km/h, reaction time 1.0 s, deceleration 5.0 m/s² to rest.

Convert: $72 \div 3.6 = 20$ m/s

Reaction phase: $20 \text{ m/s} \times 1.0 \text{ s} = 20 \text{ m}$

Braking phase: $v^2 = 2as$ with final speed 0, so $s = 20^2 \div (2 \times 5.0) = 400 \div 10 = 40 \text{ m}$.

Total: $20 + 40 = 60 \text{ m}$

Result: 60 m.

Classic slip: 40 m forgets the reaction phase, and 80 m multiplies the full 20 m/s by the whole 4.0 s braking time instead of using the average 10 m/s.

49. A train travelling at 30 m/s applies its brakes and comes to rest after covering 150 m at constant deceleration. What is the magnitude of the deceleration?

- A)** 0.20 m/s² **B)** 1.5 m/s² **C)** 3.0 m/s² **D)** 5.0 m/s² **E)** 6.0 m/s²

Answer: C

Given: $u = 30$ m/s, final speed 0, stopping distance 150 m, constant deceleration.

Formula: $v^2 = u^2 + 2as$ with $v = 0$, so $0 = 30^2 + 2 \times a \times 150$

Solve: $a = -900 \div 300 = -3.0 \text{ m/s}^2$, so the magnitude is 3.0 m/s²

Result: 3.0 m/s².

Classic slip: 6.0 m/s² drops the factor of 2, and 1.5 m/s² squares the average speed of 15 m/s instead of the initial speed.

50. A crate of mass 4.0 kg is pulled along a horizontal floor by a horizontal force of 24 N. A frictional force of 8.0 N opposes the motion. What is the acceleration of the crate?

- A)** 2.0 m/s² **B)** 4.0 m/s² **C)** 6.0 m/s² **D)** 8.0 m/s² **E)** 16 m/s²

Answer: B

Given: $m = 4.0$ kg, pull 24 N, friction 8.0 N opposing the pull.

Resultant force: $24 - 8.0 = 16 \text{ N}$

Newton's second law: $a = F \div m = 16 \div 4.0 = 4.0 \text{ m/s}^2$

Result: 4.0 m/s².

Classic slip: 6.0 m/s² divides the pull by the mass and ignores friction, and 8.0 m/s² quotes the resultant force in newtons as if it were an acceleration.

51. A block of mass 2.0 kg slides down a frictionless slope inclined at 30° to the horizontal. What is its acceleration along the slope? ($g = 10 \text{ m/s}^2$, $\sin 30^\circ = 0.50$, $\cos 30^\circ = 0.87$.)

- A)** 5.0 m/s^2 **B)** 8.7 m/s^2 **C)** 10 m/s^2 **D)** 17.4 m/s^2 **E)** 20 m/s^2

Answer: A

Given: $m = 2.0 \text{ kg}$, slope at 30° , frictionless, $g = 10 \text{ m/s}^2$, $\sin 30^\circ = 0.50$

Only the weight component along the slope accelerates the block: $F = mg \sin 30^\circ$.

Substitute: $2.0 \times 10 = 20 \text{ N}$ weight, then $20 \times 0.50 = 10 \text{ N}$ along the slope.

Newton's second law: $a = 10 \div 2.0 = 5.0 \text{ m/s}^2$, equivalently $a = g \sin 30^\circ$ since the mass cancels.

Result: 5.0 m/s^2 down the slope.

Classic slip: 8.7 m/s^2 uses $\cos 30^\circ$, the component perpendicular to the slope, and 17.4 or 20 m/s^2 quote forces in newtons as accelerations.

52. A crane lifts a 50 kg load by applying a vertical upward force of 550 N. What is the upward acceleration of the load? ($g = 10 \text{ m/s}^2$.)

- A)** 0.10 m/s^2 **B)** 1.0 m/s^2 **C)** 10 m/s^2 **D)** 11 m/s^2 **E)** 21 m/s^2

Answer: B

Given: $m = 50 \text{ kg}$, upward pull 550 N, $g = 10 \text{ m/s}^2$

Weight: $50 \times 10 = 500 \text{ N}$ downwards

Resultant upward force: $550 - 500 = 50 \text{ N}$

Newton's second law: $a = 50 \div 50 = 1.0 \text{ m/s}^2$

Result: 1.0 m/s^2 upward.

Classic slip: 11 m/s^2 ignores the weight entirely, and 21 m/s^2 adds two forces that act in opposite directions.

53. An astronaut of mass 60 kg floats at rest in space and throws a tool of mass 2.0 kg so that it moves away from her at 6.0 m/s. What is the velocity of the astronaut immediately afterwards?

- A)** 0.20 m/s in the direction of the throw.
B) 0.20 m/s opposite to the direction of the throw.
C) 0.40 m/s opposite to the direction of the throw.
D) 6.0 m/s opposite to the direction of the throw.
E) 12 m/s opposite to the direction of the throw.

Answer: B

Given: astronaut 60 kg at rest, tool 2.0 kg thrown at 6.0 m/s.

Total momentum before the throw: 0, and momentum is conserved, so it stays 0.

Tool momentum: $2.0 \times 6.0 = 12 \text{ kg}\cdot\text{m/s}$ in the throw direction

Astronaut momentum: $12 \text{ kg}\cdot\text{m/s}$ the opposite way, so her speed is $12 \div 60 = 0.20 \text{ m/s}$.

Result: 0.20 m/s opposite to the direction of the throw.

Classic slip: 12 m/s reads the momentum value as a velocity, and 6.0 m/s assumes her speed equals the tool's.

54. A ball of mass 0.50 kg strikes a wall horizontally at 12 m/s and rebounds along its original line at 8.0 m/s. The contact with the wall lasts 0.020 s. What is the magnitude of the average force exerted by the wall on the ball?

A) 100 N B) 200 N C) 300 N D) 500 N E) 1000 N

Answer: D

Given: $m = 0.50$ kg, strikes at 12 m/s, rebounds at 8.0 m/s, contact time 0.020 s.

Momentum before: $0.50 \times 12 = 6.0$ kg·m/s; momentum after: $0.50 \times 8.0 = 4.0$ kg·m/s in the opposite direction.

Because the direction reverses, $\Delta p = 6.0 + 4.0 = 10$ kg·m/s

Impulse law: $F = \Delta p \div \Delta t = 10 \div 0.020 = 500$ N

Result: 500 N.

Classic slip: 100 N subtracts the speeds as if the ball continued through the wall without rebounding, and 1000 N adds the speeds but forgets the mass.

55. A motor lifts a 60 kg load at constant speed through a height of 10 m in 20 s. What is the useful output power of the motor? ($g = 10$ m/s².)

A) 30 W B) 150 W C) 300 W D) 600 W E) 6000 W

Answer: C

Given: $m = 60$ kg lifted at constant speed through 10 m in 20 s, $g = 10$ m/s².

Constant speed: lifting force = weight = $60 \times 10 = 600$ N.

Work: $600 \times 10 = 6000$ J

Power: $6000 \div 20 = 300$ W

Result: 300 W.

Classic slip: 6000 W quotes the work in joules as if it were a power, and 30 W leaves g out of the work.

56. A car moves at a constant 20 m/s along a horizontal road. The total resistive force on it is 600 N. What power must the engine deliver to maintain this speed?

A) 30 W B) 600 W C) 6000 W D) 12000 W E) 24000 W

Answer: D

Given: constant speed 20 m/s, total resistance 600 N.

Constant speed: driving force = resistance = 600 N.

Power: $P = F \times v = 600 \times 20 = 12\,000$ W

Result: 12 000 W.

Classic slip: 30 W divides the force by the speed instead of multiplying, and 600 W is just the force itself rather than force times velocity.

Section F - Fluids, thermal physics and electricity (Q57-75)

57. A diver works at a depth of 15 m in a fresh-water lake. The density of water is 1000 kg/m^3 , $g = 10 \text{ m/s}^2$, and the atmospheric pressure at the surface is 100 kPa. Which statement about the pressures on the diver is correct?

- A)** The absolute pressure at 15 m is 150 kPa, because atmospheric pressure does not act underwater.
- B)** The gauge pressure at 15 m is 250 kPa, and the absolute pressure is 350 kPa.
- C)** The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 50 kPa.
- D)** The gauge pressure at 15 m is 1.5 kPa, and the absolute pressure is 101.5 kPa.
- E)** The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 250 kPa.

Answer: E

Given: depth 15 m fresh water, $\rho = 1000 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$, atmospheric pressure 100 kPa.

Gauge pressure: $p_{gh} = 1000 \times 10 \times 15 = 150\,000 \text{ Pa} = 150 \text{ kPa}$

Absolute pressure adds the atmosphere pressing on the surface, transmitted down through the water:
 $150 + 100 = 250 \text{ kPa}$.

Result: gauge 150 kPa and absolute 250 kPa.

Classic slip: quoting the 150 kPa as the absolute pressure forgets the atmosphere acting on the surface.

58. An open-tube manometer contains mercury of density 13600 kg/m^3 . The mercury level in the arm open to the atmosphere stands 100 mm above the level in the arm connected to a gas supply. Atmospheric pressure is 100 kPa and $g = 10 \text{ m/s}^2$. What is the absolute pressure of the gas supply?

- A)** 13.6 kPa **B)** 86.4 kPa **C)** 100 kPa **D)** 101 kPa **E)** 113.6 kPa

Answer: E

Given: mercury $\rho = 13600 \text{ kg/m}^3$, open arm 100 mm = 0.100 m above the gas arm, atmosphere 100 kPa, $g = 10 \text{ m/s}^2$.

The open arm stands higher, so the gas pressure exceeds atmospheric: $p_{\text{gas}} = p_{\text{atm}} + p_{gh}$.

Column: $13600 \times 10 = 136\,000$, then $136\,000 \times 0.100 = 13\,600 \text{ Pa} = 13.6 \text{ kPa}$

Add: $100 + 13.6 = 113.6 \text{ kPa}$

Result: 113.6 kPa.

Classic slip: 86.4 kPa subtracts the column instead of adding it, and 101 kPa computes the column with the density of water.

59. A tank contains 0.80 m of oil of density 800 kg/m^3 floating on 1.2 m of water of density 1000 kg/m^3 . Taking $g = 10 \text{ m/s}^2$ what is the gauge pressure at the bottom of the tank?

- A)** 1.8 kPa **B)** 6.4 kPa **C)** 12 kPa **D)** 18.4 kPa **E)** 20 kPa

Answer: D

Given: oil 0.80 m at 800 kg/m^3 over water 1.2 m at 1000 kg/m^3 , $g = 10 \text{ m/s}^2$

Each layer adds its own p_{gh} .

Oil: $800 \times 10 \times 0.80 = 6400 \text{ Pa} = 6.4 \text{ kPa}$

Water: $1000 \times 10 \times 1.2 = 12\,000 \text{ Pa} = 12 \text{ kPa}$

Total gauge: $6.4 + 12 = 18.4 \text{ kPa}$

Result: 18.4 kPa.

Classic slip: 20 kPa uses the density of water for both layers, and 12 kPa forgets the oil layer.

60. Three blocks are placed in oil of density 800 kg/m^3 . Block P floats with exactly half its volume submerged, block Q floats with one quarter of its volume submerged, and block R sinks and rests on the bottom. Which statement is correct?

- A)** Q has a density of about 200 kg/m^3 , the lowest of the three, and R is denser than the oil.
- B)** R has a density of exactly 800 kg/m^3 , the same as the oil.
- C)** P and Q must have the same density, because both are floating.
- D)** Q is denser than P, because a smaller fraction of Q is submerged.
- E)** P is the densest of the three blocks.

Answer: A

Given: oil 800 kg/m^3 ; P floats half submerged, Q floats one quarter submerged, R rests on the bottom.

Floating rule: a floater displaces its own weight, so block density = submerged fraction $\times$ fluid density.

$$\text{P: } 0.5 \times 800 = 400 \text{ kg/m}^3$$

$$\text{Q: } 0.25 \times 800 = 200 \text{ kg/m}^3$$

R rests on the bottom, so its density must exceed 800 kg/m^3 .

Result: Q at 200 kg/m^3 is the least dense, and R is denser than the oil.

Classic slip: calling Q denser than P inverts the rule: the smaller the submerged fraction, the lower the density.

61. A block of weight 6.0 N and density 750 kg/m^3 floats freely in water of density 1000 kg/m^3 . What is the apparent weight of the block while it floats?

- A)** 0 N **B)** 1.5 N **C)** 2.0 N **D)** 4.5 N **E)** 6.0 N

Answer: A

Given: block weight 6.0 N , density 750 kg/m^3 , floating freely in water of density 1000 kg/m^3 .

Floating means equilibrium: upthrust = weight = 6.0 N , since the block submerges the fraction $750/1000$ of its volume and displaces its own weight of water.

Net force: 6.0 N up plus 6.0 N down = 0 , so the apparent weight is 0 N .

Result: 0 N .

Classic slip: 1.5 N subtracts only three quarters of the weight, but the whole weight is already supported by the upthrust.

62. How much heat must be supplied to 2.0 kg of water to raise its temperature from 20°C to 70°C ? (c of water = 4200 J/(kg K) ; ignore heat losses.)

- A)** 84 kJ **B)** 210 kJ **C)** 294 kJ **D)** 420 kJ **E)** 588 kJ

Answer: D

Given: $m = 2.0 \text{ kg}$ of water, $c = 4200 \text{ J/(kg K)}$, heated from 20°C to 70°C

Temperature rise: $\Delta T = 70 - 20 = 50 \text{ K}$

Formula: $Q = mc\Delta T$.

Substitute: $2.0 \times 4200 \times 50$.

Steps: $2.0 \times 4200 = 8400$, then $8400 \times 50 = 420\,000 \text{ J} = 420 \text{ kJ}$

Result: 420 kJ .

Classic slip: 588 kJ uses the final temperature 70 instead of the 50 K rise, and 210 kJ corresponds to only 1.0 kg of water.

63. A 0.40 kg block of aluminium at 100 °C is dropped into 0.60 kg of water at 20 °C in an insulated container. (c of aluminium = 900 J/(kg K), c of water = 4200 J/(kg K).) What is the final equilibrium temperature?

- A)** 20 °C **B)** 30 °C **C)** 40 °C **D)** 52 °C **E)** 60 °C

Answer: B

Given: aluminium 0.40 kg at 100 °C (c = 900 J/(kg·K)) into water 0.60 kg at 20 °C (c = 4200 J/(kg·K)), insulated.

$$\text{Heat balance: } 0.40 \times 900 \times (100 - T) = 0.60 \times 4200 \times (T - 20)$$

$$\text{Left side: } 360 \times (100 - T) = 36\,000 - 360T; \text{ right side: } 2520 \times (T - 20) = 2520T - 50\,400$$

$$\text{So } 36\,000 - 360T = 2520T - 50\,400, \text{ giving } 36\,000 + 50\,400 = 2520T + 360T$$

$$86\,400 = 2880T, \text{ so } T = 86\,400 \div 2880 = 30 \text{ °C}$$

Result: 30 °C.

Classic slip: 52 °C is the mass-weighted average of the starting temperatures, ignoring that water's specific heat is far larger than aluminium's.

64. In an insulated cup, 0.50 kg of water at 12 °C is mixed with hot water at 82 °C, and the mixture settles at 32 °C. (c of water = 4200 J/(kg K); ignore heat losses.) What mass of hot water was added?

- A)** 0.10 kg **B)** 0.20 kg **C)** 0.35 kg **D)** 0.50 kg **E)** 0.70 kg

Answer: B

Given: 0.50 kg water at 12 °C mixed with m kg water at 82 °C, settling at 32 °C, same c on both sides.

Since c cancels: $m_{\text{cold}} \times \Delta T_{\text{cold}} = m_{\text{hot}} \times \Delta T_{\text{hot}}$.

Rises and falls: the cold water rises $32 - 12 = 20$ K; the hot water falls $82 - 32 = 50$ K.

$$\text{Substitute: } 0.50 \times 20 = m \times 50, \text{ so } 10 = 50m$$

$$\text{Divide: } m = 10 \div 50 = 0.20 \text{ kg}$$

Result: 0.20 kg of hot water.

Classic slip: 0.70 kg uses the 70 K gap between the two starting temperatures instead of the cold water's actual 20 K rise.

65. An electric heater consumes 2.5 kWh of energy. (1 kWh = 3.6 MJ.) This energy, expressed in megajoules, is:

- A)** 0.90 MJ **B)** 2.5 MJ **C)** 9.0 MJ **D)** 25 MJ **E)** 90 MJ

Answer: C

Given: 2.5 kWh, with 1 kWh = 3.6 MJ

$$\text{Convert: } 2.5 \times 3.6 = 9.0 \text{ MJ}$$

Result: 9.0 MJ.

Classic slip: 2.5 MJ treats 1 kWh as 1 MJ and drops the factor 3.6.

66. A heating element connected across a 12 V supply carries a current of 2.0 A. How much energy does it convert to heat in 5.0 minutes?

- A)** 0.60 kJ **B)** 0.72 kJ **C)** 3.6 kJ **D)** 7.2 kJ **E)** 72 kJ

Answer: D

Given: $V = 12 \text{ V}$, $I = 2.0 \text{ A}$, $t = 5.0 \text{ minutes}$

Power: $P = VI = 12 \times 2.0 = 24 \text{ W}$

Time in seconds: $5.0 \times 60 = 300 \text{ s}$

Energy: $E = Pt = 24 \times 300 = 7200 \text{ J} = 7.2 \text{ kJ}$

Result: 7.2 kJ.

Classic slip: 3.6 kJ multiplies voltage by time and forgets the current, and 0.72 kJ treats 5.0 minutes as 30 s.

67. Two identical 6.0Ω resistors are connected, first in series and then in parallel, across a 12 V battery of negligible internal resistance. Which one of the following statements is false?

- A)** The parallel arrangement dissipates 4 times as much total power as the series arrangement.
B) In the parallel arrangement the battery supplies 48 W.
C) In the series arrangement each resistor dissipates 6.0 W.
D) In the series arrangement the battery supplies 12 W.
E) The equivalent resistance of the parallel arrangement is 6.0Ω .

Answer: E

Given: two 6.0Ω resistors across 12 V, first in series, then in parallel; find the false statement.

Series: $R = 6.0 + 6.0 = 12 \Omega$, $I = 12 \div 12 = 1.0 \text{ A}$, battery power $= 12 \times 1.0 = 12 \text{ W}$, and each resistor dissipates $I^2R = 1.0 \times 1.0 \times 6.0 = 6.0 \text{ W}$, so both series claims are true.

Parallel: $R = (6.0 \times 6.0) \div (6.0 + 6.0) = 36 \div 12 = 3.0 \Omega$, so battery power $= 12^2 \div 3.0 = 144 \div 3.0 = 48 \text{ W}$, exactly four times 12 W, so both parallel power claims are true.

The parallel equivalent is 3.0Ω , not 6.0Ω , so that statement is false.

Result: the false statement is the one giving the parallel arrangement an equivalent resistance of 6.0Ω .

Classic slip: forgetting that two equal resistors in parallel halve the resistance rather than leaving it unchanged.

68. A 6.0Ω resistor is connected in parallel with a 3.0Ω resistor, and this pair is connected in series with a 5.0Ω resistor across a supply. What is the equivalent resistance of the whole network?

- A)** 2.0Ω **B)** 5.0Ω **C)** 7.0Ω **D)** 11Ω **E)** 14Ω

Answer: C

Given: 6.0Ω parallel with 3.0Ω , the pair in series with 5.0Ω .

Parallel pair: $(6.0 \times 3.0) \div (6.0 + 3.0) = 18 \div 9 = 2.0 \Omega$

Series add: $2.0 + 5.0 = 7.0 \Omega$

Result: 7.0Ω .

Classic slip: 14Ω adds all three as if all were in series, and 2.0Ω forgets the series resistor.

69. A battery of negligible internal resistance supplies 24 V to an $8.0\ \Omega$ resistor connected in series with a parallel combination of a $12\ \Omega$ resistor and a $6.0\ \Omega$ resistor. What current does the battery supply?

- A)** 1.5 A **B)** 2.0 A **C)** 2.4 A **D)** 3.0 A **E)** 6.0 A

Answer: B

Given: 24 V battery, $8.0\ \Omega$ in series with a parallel pair of $12\ \Omega$ and $6.0\ \Omega$.

$$\text{Parallel pair: } (12 \times 6.0) \div (12 + 6.0) = 72 \div 18 = 4.0\ \Omega$$

$$\text{Total resistance: } 8.0 + 4.0 = 12\ \Omega$$

$$\text{Current: } I = 24 \div 12 = 2.0\ \text{A}$$

Result: 2.0 A.

Classic slip: 3.0 A divides 24 V by the $8.0\ \Omega$ alone and ignores the parallel section, and 1.5 A doubles the total resistance.

70. A $6.0\ \Omega$ resistor draws 2.0 A from a battery of negligible internal resistance. When a second resistor is connected in parallel with it, the current drawn from the battery rises to 3.0 A. What is the resistance of the second resistor?

- A)** $3.0\ \Omega$ **B)** $6.0\ \Omega$ **C)** $12\ \Omega$ **D)** $18\ \Omega$ **E)** $36\ \Omega$

Answer: C

Given: a $6.0\ \Omega$ resistor draws 2.0 A; adding a parallel resistor raises the total to 3.0 A.

$$\text{Battery voltage: } V = 6.0 \times 2.0 = 12\ \text{V}$$

Parallel branches share the same 12 V, so the original branch still draws 2.0 A.

$$\text{New branch current: } 3.0 - 2.0 = 1.0\ \text{A}$$

$$\text{Its resistance: } 12 \div 1.0 = 12\ \Omega$$

Result: $12\ \Omega$.

Classic slip: $6.0\ \Omega$ assumes the extra current is shared equally between the branches, and $18\ \Omega$ adds resistances as if the second resistor were in series.

71. A $4.0\ \Omega$ resistor and a $12\ \Omega$ resistor are connected in series across a 6.0 V battery of negligible internal resistance. A wire of negligible resistance is then connected across the $12\ \Omega$ resistor. What current does the battery supply once the wire is connected?

- A)** 0.375 A **B)** 0.50 A **C)** 1.5 A **D)** 3.0 A **E)** 6.0 A

Answer: C

Given: $4.0\ \Omega$ in series with $12\ \Omega$ across 6.0 V; a wire of negligible resistance is added across the $12\ \Omega$.

The wire short-circuits the $12\ \Omega$: all the current bypasses it.

Remaining resistance: $4.0\ \Omega$ only.

$$\text{Current: } I = 6.0 \div 4.0 = 1.5\ \text{A}$$

Result: 1.5 A.

Classic slip: 0.375 A is the current before the wire is added ($6.0 \div 16$), and 0.50 A would follow from short-circuiting the $4.0\ \Omega$ instead.

72. Intensity measures how fast a wave delivers energy to a surface, per unit area of that surface. The SI unit in which intensity is expressed is:

- A)** W/m³ **B)** J/m² **C)** W/m **D)** J/s **E)** W/m²

Answer: E

Given: intensity is energy delivered per unit time per unit surface area.

Energy per second is power, measured in watts; per unit area divides by m².

Unit: W/m², equivalently J/(s·m²).

Result: W/m².

Classic slip: J/s is the watt alone with no area division, and J/m² keeps the area but loses the per-second factor.

73. A small lamp is treated as a point source radiating uniformly in all directions. At 2.0 m from the lamp the intensity of its radiation is 36 W/m². What is the intensity at 6.0 m from the lamp?

- A)** 4.0 W/m² **B)** 6.0 W/m² **C)** 12 W/m² **D)** 18 W/m² **E)** 36 W/m²

Answer: A

Given: intensity 36 W/m² at 2.0 m from a point source radiating uniformly.

A point source spreads its power over a sphere of area $4\pi r^2$, so intensity follows an inverse-square law, $I \propto 1/r^2$.

Distance triples, from 2.0 m to 6.0 m, a factor of 3, so the intensity divides by $3^2 = 9$.

Compute: $36 \div 9 = 4.0 \text{ W/m}^2$

Result: 4.0 W/m².

Classic slip: 12 W/m² scales as $1/r$ instead of $1/r^2$.

74. In a heated room, a radiator warms the air beside it. This air rises, cooler air replaces it, and the resulting circulating flow carries thermal energy around the room. This circulating flow transports the energy mainly by:

- A)** conduction through the metal of the radiator, the best heat conductor present
B) evaporation, because the flow carries water vapour that releases heat where it condenses
C) radiation, because the hot radiator emits infrared rays that travel in straight lines
D) conduction, because neighbouring air molecules pass energy on by direct collisions
E) convection, because the warmed air expands, becomes less dense and rises within the flow

Answer: E

Givens: a radiator warms the air beside it, the warm air rises, cooler air replaces it, and the circulating flow carries the energy.

Energy transported by the bulk motion of a warmed fluid is convection by definition.

The mechanism in the stem matches exactly: heated air expands, becomes less dense, rises, and cooler air sinks to replace it.

Conduction passes energy between neighbouring molecules with no bulk flow, and radiation travels as electromagnetic waves, not as moving air.

Result: convection.

Classic slip: crediting the direct molecule-to-molecule collisions of conduction ignores the bulk circulation the stem describes.

75. A vacuum flask keeps a drink hot. It has double walls with silvered surfaces and a vacuum between them, closed at the top by a stopper. Which statement about how the flask limits heat loss is correct?

- A)** The silvered walls mainly reduce conduction through the walls.
- B)** The vacuum blocks convection only, because conduction can still occur through a vacuum.
- C)** The stopper works mainly by reflecting radiation back into the drink.
- D)** The vacuum mainly blocks radiation, and the silvering mainly blocks convection.
- E)** The vacuum blocks both conduction and convection, and the silvered surfaces cut radiation losses.

Answer: E

Givens: vacuum flask with double silvered walls, vacuum between them, stopper on top.

Conduction needs a medium and convection needs a moving fluid, so the vacuum removes both paths at once.

Radiation is infrared, and the silvered surfaces reflect it back towards the drink, cutting radiation losses.

The stopper seals the last convection route at the neck.

Result: the vacuum blocks both conduction and convection, and the silvered surfaces cut radiation losses.

Classic slip: the claim that the vacuum blocks only convection forgets that conduction has no medium in a vacuum either, and it is the silvering that handles radiation.

Section G - Logic and argument analysis (Q76-95)

76. In a library, every manuscript in the special archive is catalogued. Some of the archive's manuscripts are illustrated, and no illustrated item in the archive may leave the building. Which of the following must be true?

- A)** Some of the archive's illustrated manuscripts are catalogued.
- B)** Every catalogued item in the library is illustrated.
- C)** Every item that may leave the building is catalogued.
- D)** Some catalogued items in the library may leave the building.
- E)** No catalogued item in the archive is illustrated.

Answer: A

Givens: (1) every archive manuscript is catalogued; (2) some archive manuscripts are illustrated; (3) no illustrated archive item may leave the building.

Premise 2 guarantees at least one illustrated archive manuscript exists.

Premise 1 applies to every archive manuscript, so that same manuscript is catalogued.

So at least one manuscript is both illustrated and catalogued.

Result: some of the archive's illustrated manuscripts are catalogued.

Classic slip: every catalogued item being illustrated is the invalid converse of premise 1, and items that may leave need not be catalogued because non-archive items are unrestricted.

77. At a tennis club, only members with a current season pass may book the courts, and all club rules are followed. Nadia booked a court this week. Some members with a season pass also take coaching lessons. Which of the following must be true?

- A)** Nadia holds a current season pass.
- B)** Every member who books the courts takes coaching lessons.
- C)** Some members who book the courts take coaching lessons.
- D)** Every member with a season pass books the courts.
- E)** Nadia takes coaching lessons.

Answer: A

Givens: (1) only members with a current season pass may book the courts; (2) Nadia booked a court; (3) some pass holders take coaching lessons.

The rule 'only pass holders may book' decodes as: booking the courts implies holding a current season pass.

Nadia booked a court, so by that rule she holds a current season pass.

Premise 3 concerns some pass holders, who need not be Nadia.

Result: Nadia holds a current season pass.

Classic slip: the claim that every pass holder books the courts merely reverses the rule, and the lesson statements concern pass holders in general.

78. Every swimmer who won a medal this season trains at the Riverside pool, and nobody who trains at the Riverside pool pays a joining fee. Which of the following must be true?

- A)** Some people who train at the Riverside pool did not win a medal this season.
- B)** The Riverside pool charges no joining fee to anyone.
- C)** Some swimmers who pay a joining fee won a medal this season.
- D)** Every swimmer who trains at the Riverside pool won a medal this season.
- E)** No swimmer who won a medal this season pays a joining fee.

Answer: E

Givens: (1) every medalling swimmer trains at Riverside; (2) nobody who trains at Riverside pays a joining fee.

Chain the universals: medallist → trains at Riverside → pays no joining fee.

So any medal winner pays no joining fee.

The premises say nothing forcing Riverside trainers to have medalled, and nothing about fees for people who do not train at Riverside.

Result: no swimmer who won a medal this season pays a joining fee.

Classic slip: the claim that every Riverside trainer medalled reverses the direction of the first premise.

79. Some players in the squad are left-footed. No left-footed player takes the corner kicks, and every player in the squad is a professional. Which of the following must be true?

- A)** Some professionals do not take the corner kicks.
- B)** Every player who takes the corner kicks is right-footed.
- C)** No professional takes the corner kicks.
- D)** Every player who takes the corner kicks is in the squad.
- E)** Some left-footed players are not professionals.

Answer: A

Givens: (1) some squad players are left-footed; (2) no left-footed player takes the corners; (3) every squad player is a professional.

Premise 1 gives at least one left-footed squad player.

Premise 3 makes that player a professional.

Premise 2 says that same player does not take the corners.

So at least one professional does not take the corners.

Result: some professionals do not take the corner kicks.

Classic slip: the claim that every corner taker is right-footed assumes that not left-footed means right-footed, which no premise states.

80. Every baker at the Saturday market arrives before dawn, and some bakers at the market sell rye bread. Which of the following must be true?

- A)** Everyone who arrives before dawn sells rye bread.
- B)** Some people who arrive before dawn sell rye bread.
- C)** All sellers of rye bread at the market are bakers.
- D)** No baker at the market sells rye bread.
- E)** Some bakers at the market arrive after dawn.

Answer: B

Givens: (1) every baker at the market arrives before dawn; (2) some bakers at the market sell rye bread.

Premise 2 gives at least one rye-selling baker.

Premise 1 makes that same baker a before-dawn arrival.

So at least one before-dawn arrival sells rye bread.

Result: some people who arrive before dawn sell rye bread.

Classic slip: the claim that everyone arriving before dawn sells rye reverses the first premise, and nothing makes rye sellers bakers in general.

81. Every member of the rowing club can swim, and no member of the rowing club is under sixteen. Tara is a member of the rowing club. Which of the following must be true?

- A)** Some people under sixteen can swim.
- B)** Everyone who can swim is a member of the rowing club.
- C)** Tara is under sixteen.
- D)** Only swimmers may join the rowing club.
- E)** Tara can swim and is not under sixteen.

Answer: E

Givens: (1) every member of the rowing club can swim; (2) no member is under sixteen; (3) Tara is a member.

A universal premise applies to every member, Tara included.

From 1: Tara can swim.

From 2: Tara is not under sixteen.

Result: Tara can swim and is not under sixteen.

Classic slip: the claim that everyone who can swim is a member reverses the first premise, and the premises say nothing about who is allowed to join.

82. About a postponed football match, Ria says: 'The match was postponed because of rain.' Tom says: 'The match was postponed because of rain, and Ria is lying.' Exactly one of the two always tells the truth and the other always lies. Which of the following must be true?

- A)** Tom tells the truth and rain caused the postponement.
- B)** Ria tells the truth and rain caused the postponement.
- C)** Ria tells the truth and the postponement had another cause.
- D)** Tom tells the truth and the postponement had another cause.
- E)** Both of them always tell the truth.

Answer: B

Givens: Ria claims rain caused the postponement; Tom claims rain caused it and Ria is lying; exactly one of them always tells the truth.

Test Tom as the truth teller: both halves of his claim hold, so rain caused it and Ria is lying.

But Ria's claim is exactly that rain caused it, so Ria lying would make rain not the cause, contradicting Tom's first half.

Hence Tom is the liar and Ria the truth teller.

Ria truthful means rain did cause the postponement, and Tom's conjunction fails on its second half.

Result: Ria tells the truth and rain caused the postponement.

Classic slip: the branch with Tom truthful and rain the cause is exactly the one that produced the contradiction above.

83. Three pupils, Amara, Ben and Cara, are asked about a missing lab key. Each of them either always tells the truth or always lies. Amara says: 'Ben is lying.' Ben says: 'Cara is lying.' Cara says: 'Both Amara and Ben are lying.' Which of the following is correct?

- A)** Amara and Cara lie and Ben tells the truth.
- B)** The three statements allow more than one consistent assignment.
- C)** Ben and Cara lie and Amara tells the truth.
- D)** Amara and Ben tell the truth and Cara lies.
- E)** All three tell the truth.

Answer: A

Givens: Amara says Ben lies; Ben says Cara lies; Cara says both Amara and Ben lie; each person is all-truth or all-lie.

Test Amara truthful: then Ben lies, so Ben's claim that Cara lies is false and Cara is truthful.

But Cara's claim that both Amara and Ben lie is false, since Amara is truthful, impossible for a truth teller.

So Amara lies: her claim that Ben lies is false, so Ben is truthful.

Ben truthful means Cara lies, and Cara's claim that both Amara and Ben lie is indeed false, consistent for a liar.

Result: Amara and Cara lie, and Ben tells the truth.

Classic slip: the assignment is unique; every alternative branch hits one of the contradictions above.

84. Three teachers are questioned about a borrowed stopwatch. Exactly one of them is lying and the other two are telling the truth. Priya says: 'I did not take the stopwatch.' Quinn says: 'Priya is lying.' Ravi says: 'Quinn is lying.' Which of the following must be true?

- A)** Priya took the stopwatch.
- B)** All three are telling the truth.
- C)** Quinn took the stopwatch.
- D)** Ravi took the stopwatch.
- E)** Priya did not take the stopwatch.

Answer: E

Givens: Priya says she did not take the stopwatch; Quinn says Priya is lying; Ravi says Quinn is lying; exactly one of the three is lying.

Quinn and Ravi make opposite claims, so exactly one of that pair must be lying whichever way the facts go.

The single liar therefore sits in the Quinn-Ravi pair, so Priya is truthful and her denial is accurate.

Since Priya is truthful, Quinn's claim that she is lying is false, so Quinn is the liar and Ravi tells the truth.

Result: Priya did not take the stopwatch.

Classic slip: naming any culprit overreaches, since the statements fix only Priya's innocence, not who is guilty.

85. After a match, the scoreboard was found broken. Exactly one of the three candidates for team captain always tells the truth and the other two always lie. Dana says: 'Eli broke the scoreboard.' Eli says: 'I did not break the scoreboard.' Fran says: 'Dana is lying.' Which of the following must be true?

- A)** Eli tells the truth and Fran broke the scoreboard.
- B)** Fran tells the truth and Dana broke the scoreboard.
- C)** Eli tells the truth and Dana broke the scoreboard.
- D)** Dana tells the truth and Eli broke the scoreboard.
- E)** Fran tells the truth and Eli broke the scoreboard.

Answer: D

Givens: Dana says Eli broke the scoreboard; Eli says he did not; Fran says Dana is lying; exactly one truth teller.

Test Eli truthful: then Dana and Fran both lie, but Fran's claim that Dana is lying would then be true, impossible for a liar.

Test Fran truthful: then Dana lies, so Eli did not break it, which makes Eli's denial true as well, giving two truth tellers, impossible.

So Dana is the single truth teller: her claim holds, Eli broke it, and Eli's denial is his lie; Fran's claim that Dana lies is also false, consistent for a liar.

Result: Dana tells the truth and Eli broke the scoreboard.

Classic slip: any branch with Eli or Fran truthful collapses by one of the two contradictions above.

86. At a lost property office, exactly one of Pia's two statements is true and exactly one of Sam's two statements is true. The clerk confirms the blue bag was collected on exactly one day. Pia says: 'The bag was collected on Monday.' and 'The bag was not collected on Wednesday.' Sam says: 'The bag was collected on Wednesday.' and 'The bag was not collected on Monday.' Which of the following must be true?

- A)** The bag was collected on Wednesday.
- B)** The bag was collected on Monday.
- C)** Sam made two true statements.
- D)** Pia made two false statements.
- E)** The bag was collected on a day that is neither Monday nor Wednesday.

Answer: E

Givens: Pia says 'collected on Monday' and 'not collected on Wednesday', exactly one of hers true; Sam says 'collected on Wednesday' and 'not collected on Monday', exactly one of his true; the collection happened on exactly one day.

Test Monday: Pia's 'Monday' is true and 'not Wednesday' is also true, two true statements, violating her exactly-one rule.

Test Wednesday: Pia's 'Monday' is false and 'not Wednesday' is false too, zero true, again violating her rule.

Any other day: 'not Wednesday' is true and 'Monday' false, exactly one true for Pia, and symmetrically exactly one true for Sam.

Result: the bag was collected on a day that is neither Monday nor Wednesday.

Classic slip: Monday and Wednesday each break the exactly-one condition rather than satisfy it.

87. A notice at the swimming pool reads: 'If the pool is at capacity, new swimmers are turned away.' Which of the following is logically equivalent to the notice?

- A)** If the pool is not at capacity, new swimmers are turned away.
- B)** If new swimmers are not turned away, the pool is not at capacity.
- C)** If new swimmers are turned away, the pool is at capacity.
- D)** If the pool is not at capacity, new swimmers are not turned away.
- E)** New swimmers are turned away only if the pool is at capacity.

Answer: B

Givens: the notice reads 'at capacity $\rightarrow$ new swimmers turned away'.

A conditional is equivalent only to its contrapositive: negate both parts and swap them.

Contrapositive: not turned away $\rightarrow$ not at capacity.

The converse 'turned away $\rightarrow$ at capacity' and the inverse 'not at capacity $\rightarrow$ not turned away' both fail.

Result: if new swimmers are not turned away, the pool is not at capacity.

Classic slip: reading the notice backwards treats being turned away as proof of capacity, but a pool below capacity may still turn people away for other reasons.

88. The coach tells the squad: 'If a player attends every training session this month, she will be selected for the final.' Which of the following is logically equivalent to the coach's statement?

- A)** A player who attends at least one training session this month will be selected for the final.
- B)** A player will be selected for the final only if she attends every training session this month.
- C)** If a player does not attend every training session this month, she will not be selected for the final.
- D)** If a player is not selected for the final, she did not attend every training session this month.
- E)** If a player is selected for the final, she attended every training session this month.

Answer: D

Givens: the coach's promise reads 'perfect attendance $\rightarrow$ selected for the final'.

Only the contrapositive is equivalent: negate both parts and swap them.

Contrapositive: not selected $\rightarrow$ attendance was not perfect.

The inverse 'imperfect attendance $\rightarrow$ not selected' and the converse 'selected $\rightarrow$ perfect attendance' do not follow.

Result: if a player is not selected for the final, she did not attend every training session this month.

Classic slip: the inverse would fail exactly when a player misses a session yet is selected anyway, which the promise never rules out.

89. Under the library's loan rule, a book may be borrowed for a full term only if it is on the reserve list. Which of the following must also be true if the rule is obeyed?

- A)** A book not on the reserve list may not be borrowed for a full term.
- B)** Every book on the reserve list is borrowed for a full term.
- C)** A book not on the reserve list may be borrowed for a full term.
- D)** A book on the reserve list may be borrowed for a full term.
- E)** Only books on the reserve list may be borrowed at all.

Answer: A

Givens: the rule 'full term only if on the reserve list' decodes as full-term borrowing $\rightarrow$ reserve listed.

Equivalent form: the contrapositive, not reserve listed $\rightarrow$ no full-term borrowing.

The converse 'reserve listed $\rightarrow$ full term allowed' overstates the rule, and nothing bans shorter loans.

Result: a book not on the reserve list may not be borrowed for a full term.

Classic slip: reading 'only if' backwards treats a reserve listing as a permission slip, which the rule never grants.

90. A ferry company's policy states that if a sailing is delayed, meal vouchers are issued, and that if meal vouchers are issued, the terminal canteen stays open late. Which of the following is logically equivalent to the policy taken as a whole?

- A)** Every delayed sailing leaves passengers hungry.
- B)** If a sailing is delayed, it is because the canteen plans to stay open late.
- C)** If the canteen does not stay open late, the sailing was not delayed.
- D)** If a sailing is not delayed, the canteen does not stay open late.
- E)** If the canteen stays open late, the sailing was delayed.

Answer: C

Givens: the policy reads delayed $\rightarrow$ vouchers issued, and vouchers issued $\rightarrow$ canteen open late.

Chain: delayed $\rightarrow$ canteen open late.

Contrapositive of the chain: canteen not open late $\rightarrow$ not delayed.

Statements starting from 'not delayed', or inferring a delay from a late canteen, run the chain backwards and are not entailed.

Result: if the canteen does not stay open late, the sailing was not delayed.

Classic slip: the version 'late canteen $\rightarrow$ delayed' reverses the implication, since the policy promises vouchers after a delay but never says a delay is the only route to a late canteen.

91. A security manual states: 'The side gate stays locked if and only if the alarm is armed.' Which of the following is logically equivalent to this statement?

- A)** The alarm is armed whenever the side gate is unlocked.
- B)** If the alarm is disarmed, the side gate is locked.
- C)** If the alarm is armed, the side gate is unlocked.
- D)** If the side gate is locked, the alarm is disarmed.
- E)** The side gate is unlocked exactly when the alarm is disarmed.

Answer: E

Givens: the manual reads gate locked if and only if alarm armed.

A biconditional holds in both matching states: locked with armed, unlocked with disarmed.

Negating both sides together preserves it: unlocked exactly when disarmed.

Pairing unlocked with armed, or locked with disarmed, breaks the two-way match and contradicts the manual.

Result: the side gate is unlocked exactly when the alarm is disarmed.

Classic slip: mixing a negated side with the plain other side turns the equivalence into its contradiction.

92. A council reports that streets repaved with a new porous surface drain rainwater quickly and form far fewer puddles, and concludes that repaving will reduce road accidents in which vehicles slip. The conclusion depends on assuming which of the following?

- A)** Accidents in which vehicles slip are usually caused by surface water, and the porous surface does not make roads slippery in other ways.
- B)** Drivers on repaved streets tend to be more cautious than other drivers.
- C)** The porous surface costs less per kilometre than standard asphalt.
- D)** The council intends to repave most streets within five years.
- E)** Standard asphalt forms more puddles than the porous surface.

Answer: A

Givens: porous surface → quick drainage and far fewer puddles; conclusion, repaving → fewer slip accidents.

Gap 1: fewer puddles cuts slips only if surface water is what usually makes vehicles slip.

Gap 2: the new surface must not add slipperiness of its own in other ways.

The statement combining exactly those two ideas bridges the argument.

Result: the needed assumption is that slip accidents are usually caused by surface water and the porous surface introduces no new slipperiness.

Classic slip: the asphalt comparison merely restates a premise, and driver caution would be a rival explanation rather than a needed assumption.

93. At a gym, members on the flexible plan attend twice as often as members on the standard plan and report larger fitness gains. The manager concludes that the flexible plan itself causes the better fitness. Which of the following, if true, most weakens the manager's conclusion?

- A)** The flexible plan costs more per month than the standard plan.
- B)** Members choose their own plan, and most members who were already training hard before joining picked the flexible plan.
- C)** Standard plan members report busier weekday schedules.
- D)** Attendance at the gym is recorded automatically at the entrance.
- E)** Both groups took the same fitness assessment at the start and at the end of the study.

Answer: B

Givens: flexible-plan members attend twice as often and report larger gains; the manager concludes the plan causes the better fitness.

Alternative cause: if members choose their own plans, members already training hard may have picked the flexible plan before the study began.

Then their attendance and gains predate the plan, so the plan need not cause anything; this is self-selection.

The higher fee and the automatic recording do not touch the causal direction, and the identical assessments strengthen the study.

Result: the weakening statement is that members choose their own plan and most members already training hard picked the flexible plan.

Classic slip: the shared assessments make the comparison fairer, so they cannot weaken the conclusion.

94. A ferry operator installed automatic ticket scanners on two of its six routes. Boarding time on those two routes fell by a third within two months and stayed flat on the other four. The operator concludes the scanners caused the faster boarding. Which of the following, if true, most strengthens this conclusion?

- A)** Passengers on the scanner routes said in a survey that they liked the scanners.
- B)** Both scanner routes use ferries of the same model.
- C)** Over the same two months the operator ran an advertising campaign on all six routes.
- D)** When the scanners were out of service for two days, boarding times on those routes briefly returned to their old level and fell again after repairs.
- E)** Staff numbers were also increased at the terminals of the two scanner routes.

Answer: D

Givens: scanners on two of six routes; boarding time fell by a third on those two and stayed flat on the other four; conclusion, the scanners caused it.

Strongest support should reconnect effect to cause when the cause is removed and restored.

With the scanners out of service for two days, boarding returned to its old level and fell again after repairs, so the effect tracks the cause directly.

Extra staff on the same routes suggest a rival cause, and the advertising campaign covered all six routes, so it cannot explain the difference between them.

Result: the strengthening statement is the two-day outage that briefly restored the old boarding times.

Classic slip: the staff increase weakens the causal claim by offering an alternative explanation.

95. Last year a sports centre cut its evening swim sessions from five per week to three. Over the year, total attendance at evening swims stayed almost unchanged, and the waiting list for evening swims grew longer. Which of the following is most strongly supported by this information?

- A)** Some swimmers gave up swimming in the evenings.
- B)** The centre should restore the two cancelled sessions.
- C)** On average, each remaining evening session now accommodates more swimmers than before.
- D)** The waiting list caused the sessions to be cut.
- E)** Attendance at morning swims must have increased.

Answer: C

Givens: evening sessions cut from five to three per week; total evening attendance roughly unchanged; the waiting list for evening swims grew.

Average per session = total attendance ÷ number of sessions.

Before: total ÷ 5 per session; now: the same total ÷ 3 per session.

Since total ÷ 3 is greater than total ÷ 5, the average number of swimmers per remaining session has risen.

Result: on average each remaining evening session now accommodates more swimmers than before.

Classic slip: the recommendation to restore sessions, any change at morning sessions, and the timing of the waiting list all go beyond the data.